

WirelessMathBench

A Mathematical-Modeling Benchmark for LLMs in Wireless Communications

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The Problem

- LLMs excel at **language & code** yet struggle with **symbolic reasoning**
- 6G research demands **exact mathematics derivations**: path-loss models, MIMO channel matrices, RIS phases configurations
- **No existing benchmark** couples physical constraints with mathematical reasoning

The Mathematical Reasoning Challenge in Wireless

The Problem

- LLMs excel at **language & code** yet struggle with **symbolic reasoning**
- 6G research demands **exact mathematics derivations**: path-loss models, MIMO channel matrices, RIS phases configurations
- **No existing benchmark** couples physical constraints with mathematical reasoning

Our Solution

A specialized benchmark testing LLMs on **physics-aware mathematical modeling** for wireless communications

Positioning in the Benchmark Landscape

Existing Limitations

Current math benchmarks lack **engineering constraints** and **domain expertise**

Benchmark	Diffuculty Level	Real Engineering	Test Size
GSM8K	Elementary	✗	1.3k
MATH	High School	✗	5k
OCWCourses	University	✗	272
MMMU	University	✗	1.9k
OlympiadBench	Competition	✗	8.5k
WirelessMathBench	Expert	✓	587

Unique Value

Only benchmark with real engineering & expert level tasks

Our Contributions

1. **WirelessMathBench**: **587** expert-level questions from **40** state-of-the-art papers
2. **Multi-tier Evaluation**: MCQ $\rightarrow$ partial $\rightarrow$ full derivations for fine-grained analysis
3. **Comprehensive Study**: First systematic evaluation of **16** leading LLMs on physics-aware math tasks
4. **Open Platform**: Public leaderboard & toolkit for rapid research iteration

 <https://lixin.ai/WirelessMathBench>

Core Principles

1. Real-World Complexity

Tasks extracted from peer-reviewed wireless research papers

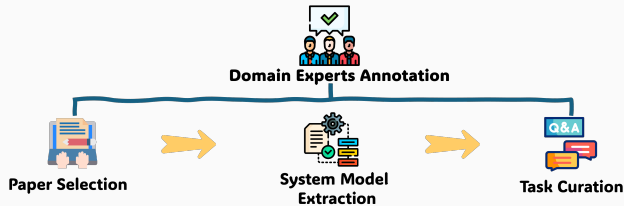
2. Progressive Difficulty

Multi-tier structure ensures fine-grained diagnostic capability

3. Physical Constraints

Every question maintains dimensional validity & engineering relevance

Dataset Construction Pipeline



Key Statistics

- **587** carefully curated Q&A pairs
- **3** progressive task types
- **40** source papers (2021–2025)
- Each question is **accurate**, **symbolically consistent**, and **self-contained**

Quality Assurance

- Domain-expert annotation
- Independent peer review

Dataset Construction – Paper Selection

Selection criteria

- **Top-tier, open-access** (published in Top Wireless Journals and open-source on arXiv)
- **Symbolic depth** – equations, not heuristics
- **Diverse coverage**: 7 scenarios, 6 problems

Data set spans **40** papers (2021–2025).

Research Areas

Category	#Papers
RIS	19
Massive MIMO	12
UAV	6
ISAC	6
Satellite	4
SIM/NOMA	5

Hybrid Workflow

1. LLM template scans latex $\rightarrow$ auto-extracts key equations.
2. Draft summarises *channel model, variables, assumptions*.
3. Domain experts refactor & verify symbolic consistency.

Two-pass review ensures every derivation is sound & unambiguous.



Task Progression: From MCQ to Full Derivation

≡ Hierarchical Structure

1. MCQ

Unit checks & dimensional consistency

2. Masked Fill (L1–L3)

Progressive symbolic reasoning

3. FEC

Full symbolic equation completion

Reasoning depth increases step-by-step—from recall to full derivation.



Wireless Channel Reflection Modeling

Background

In a double-RIS-assisted massive MIMO system, the overall channel from user k , denoted h_k , incorporates single-reflection links via RIS₁ and RIS₂, as well as a double-reflection link through both RISs. Here, N_j represents the channel from RIS_j to the BS, and D denotes the channel between the RISs. The reflection coefficients of RIS_j are given by θ_j .

Multiple Choice Question

$$h_k = [MASK]$$

Question: Which expression correctly represents the user- k effective channel at the base station?

Options:

- (A) $N_2 \text{diag}(h_{k2}) \theta_2 + N_1 \text{diag}(h_{k1}) \theta_1 + N_2 \text{diag}(\theta_2) D \text{diag}(\theta_1) h_{k1}$.
(B) $N_2 h_{k2} \theta_2 + N_1 \text{diag}(h_{k1}) \theta_1 + D \text{diag}(\theta_2) \theta_1$
(C) $N_2 \text{diag}(h_{k2}) \theta_2 + N_1 h_{k1} \theta_1 + N_2 \text{diag}(\theta_2) D \text{diag}(\theta_1) h_{k1}$
(D) $N_2 \text{diag}(h_{k2}) \theta_2 + N_1 \text{diag}(h_{k1}) \theta_1 + N_2 \text{diag}(\theta_2) D \text{diag}(\theta_1) h_{k1}$

Answer: The correct expression is [A].



Progressive Masking Fill-in-the-blank (Level 1, 2, 3)

L1

$$h_k = N_2 \text{diag}(h_{k2}) [MASK] + N_1 \text{diag}(h_{k1}) \theta_1 + N_2 \text{diag}(\theta_2) D \text{diag}(\theta_1) h_{k1}$$

Question 1: Which reflection vector is missing in the first single-reflection term?

L2

$$h_k = [MASK] \text{diag}(h_{k2}) [MASK] + N_1 \text{diag}(h_{k1}) \theta_1 + N_2 \text{diag}(\theta_2) D \text{diag}(\theta_1) h_{k1}$$

Question 2: Fill in the two interdependent channel-related variables for the RIS₂ path.

L3

$$h_k = [MASK] \text{diag}(h_{k2}) [MASK] + [MASK] \text{diag}(h_{k1}) \theta_1 + N_2 \text{diag}(\theta_2) D \text{diag}(\theta_1) h_{k1}$$

Question 3: Complete the three missing terms describing the single-reflection link via RIS₁.

Mask Ratio

$$\text{Answer: } h_k = N_2 \text{diag}(h_{k2}) \theta_2 + N_1 \text{diag}(h_{k1}) \theta_1 + N_2 \text{diag}(\theta_2) D \text{diag}(\theta_1) h_{k1}$$



Full Equation Completion

$$h_k = [MASK]$$

Question: Write the full expression for the overall effective channel.

$$\text{Answer: } h_k = N_2 \text{diag}(h_{k2}) \theta_2 + N_1 \text{diag}(h_{k1}) \theta_1 + N_2 \text{diag}(\theta_2) D \text{diag}(\theta_1) h_{k1}$$

Stage 1: MCQ — Sanity Checks

✓ Conceptual Recall

Objective: Ensure basic understanding through:

- Dimensional and unit validity
- Elimination of implausible options
- Testing surface-level knowledge

Serves as a lightweight entry point into derivation-based reasoning.



Background

In a double-RIS-assisted massive MIMO system, the overall channel from user k , denoted $\mathbf{h}_k$, incorporates single-reflection links via RIS₁ and RIS₂, as well as a double-reflection link through both RISs. Here, $\mathbf{N}_j$ represents the channel from RIS_j to the BS, and $\mathbf{D}$ denotes the channel between the RISs. The reflection coefficients of RIS_j are given by $\boldsymbol{\theta}_j$.



Multiple Choice Question

$$\mathbf{h}_k = [\mathbf{M} \mathbf{A} \mathbf{S} \mathbf{K}]$$

Question: Which expression correctly represents the user- k effective channel at the base station?

Options:



(A) $\mathbf{N}_2 \text{diag}(\mathbf{h}_{k2}) \boldsymbol{\theta}_2 + \mathbf{N}_1 \text{diag}(\mathbf{h}_{k1}) \boldsymbol{\theta}_1 + \mathbf{N}_2 \text{diag}(\boldsymbol{\theta}_2) \mathbf{D} \text{diag}(\boldsymbol{\theta}_1) \mathbf{h}_{k1}$.

(B) $\mathbf{N}_2 \mathbf{h}_{k2} \boldsymbol{\theta}_2 + \mathbf{N}_1 \text{diag}(\mathbf{h}_{k1}) \boldsymbol{\theta}_1 + \mathbf{D} \text{diag}(\boldsymbol{\theta}_2) \boldsymbol{\theta}_1$

(C) $\mathbf{N}_2 \text{diag}(\mathbf{h}_{k2}) \boldsymbol{\theta}_2 + \mathbf{N}_1 \mathbf{h}_{k1} \boldsymbol{\theta}_1 + \mathbf{N}_2 \text{diag}(\boldsymbol{\theta}_2) \mathbf{D} \text{diag}(\boldsymbol{\theta}_1) \mathbf{h}_{k1}$

(D) $\mathbf{N}_2 \text{diag}(\mathbf{h}_{k2}) \boldsymbol{\theta}_2 + \mathbf{N}_1 \text{diag}(\mathbf{h}_{k1}) \boldsymbol{\theta}_1 + \mathbf{N}_2 \text{diag}(\boldsymbol{\theta}_2) \mathbf{D} \text{diag}(\boldsymbol{\theta}_1) \mathbf{h}_{k1}$

Answer: The correct expression is .

Stage 2: Masked Fill — Guided Derivation

Symbolic Completion (L1–L3)

Objective: Evaluate structured reasoning by:

- Filling masked terms within equations
- Interpreting surrounding context
- Scaling difficulty across three levels

Tests local derivation skills without requiring full equation construction.



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Progressive Masking Fill-in-the-blank (Level 1, 2, 3)

L1

$$\mathbf{h}_k = \mathbf{N}_2 \text{diag}(\mathbf{h}_{k2}) [\text{MASK}] + \mathbf{N}_1 \text{diag}(\mathbf{h}_{k1}) \boldsymbol{\theta}_1 + \mathbf{N}_2 \text{diag}(\boldsymbol{\theta}_2) \mathbf{D} \text{diag}(\boldsymbol{\theta}_1) \mathbf{h}_{k1}$$

Question 1: Which reflection vector is missing in the first single-reflection term?

L2

$$\mathbf{h}_k = [\text{MASK}] \text{diag}(\mathbf{h}_{k2}) [\text{MASK}] + \mathbf{N}_1 \text{diag}(\mathbf{h}_{k1}) \boldsymbol{\theta}_1 + \mathbf{N}_2 \text{diag}(\boldsymbol{\theta}_2) \mathbf{D} \text{diag}(\boldsymbol{\theta}_1) \mathbf{h}_{k1}$$

Question 2: Fill in the two interdependent channel-related variables for the RIS₂ path.

L3

$$\mathbf{h}_k = [\text{MASK}] \text{diag}(\mathbf{h}_{k2}) [\text{MASK}] + [\text{MASK}] \text{diag}(\mathbf{h}_{k1}) \boldsymbol{\theta}_1 + \mathbf{N}_2 \text{diag}(\boldsymbol{\theta}_2) \mathbf{D} \text{diag}(\boldsymbol{\theta}_1) \mathbf{h}_{k1}$$

Question 3: Complete the three missing terms describing the single-reflection link via RIS₁.

Mask Ratio

$$\text{Answer: } \mathbf{h}_k = \mathbf{N}_2 \text{diag}(\mathbf{h}_{k2}) \boldsymbol{\theta}_2 + \mathbf{N}_1 \text{diag}(\mathbf{h}_{k1}) \boldsymbol{\theta}_1 + \mathbf{N}_2 \text{diag}(\boldsymbol{\theta}_2) \mathbf{D} \text{diag}(\boldsymbol{\theta}_1) \mathbf{h}_{k1}$$

Stage 3: FEC — Full Symbolic Reconstruction

Complete Derivation

Objective: Assess full modeling capabilities:

- Derive full symbolic expression from scratch
- No intermediate guidance or partial cues
- Reflect understanding of physical and logical structure

The most demanding task—measuring global symbolic synthesis ability.



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Full Equation Completion

$$\mathbf{h}_k = [MASK]$$



Question: Write the full expression for the overall effective channel.

Answer: $\mathbf{h}_k = \mathbf{N}_2 \text{diag}(\boldsymbol{\theta}_2) \boldsymbol{\theta}_2 + \mathbf{N}_1 \text{diag}(\boldsymbol{\theta}_1) \boldsymbol{\theta}_1 + \mathbf{N}_2 \text{diag}(\boldsymbol{\theta}_2) \mathbf{D} \text{diag}(\boldsymbol{\theta}_1) \mathbf{h}_{k1}$

Experimental Setup

- **16 State-of-the-Art LLMs**

DeepSeek-R1, OpenAI-o1, GPT-4o, Gemini-2, Qwen-Math, etc.

- **Standardized Evaluation**

Zero-shot prompting, unified templates

- **Robust Assessment**

GPT-4o judge for open-ended responses

Headline Results

Model	MCQ	L1	L2	L3	FEC	Avg.
DeepSeek-R1	76	60	35	12.5	7.8	38.1
OpenAI-o1	66	59	32	8.0	7.0	34.6
GPT-4o	73	43	29	6.3	4.4	30.9
Gemini-2	71	41	24	5.4	4.3	29.2
Qwen-72B	70	38	26	7.1	6.1	29.4
LLaMA-3-8B	46	11	8	2.7	2.6	13.9

Critical Insight

Even best model (DeepSeek-R1): **76%** (MCQ) → **7.8%** (FEC)

Reasoning-oriented models outperform large baselines

DeepSeek-R1, OpenAI-o1 ≫ GPT-4o, Gemini, Qwen—even with similar scale

→ *Symbolic derivation remains hard; reasoning ability matters more than model size*

Error Analysis: Why Do LLMs Fail?

🔧 Major Error Types

1. Symbol Drift

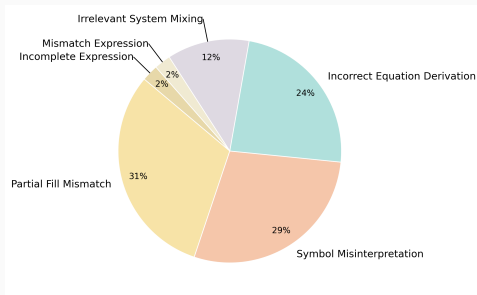
Frequent mix-ups between Hermitian and standard transpose, or wrong variable selection.

2. Chain Collapse

Early mistakes in partial masks propagate, causing irrecoverable full derivation failure.

3. Context Hallucination

Model introduces unrelated wireless concepts (e.g., NOMA) that do not fit the physical context.



Next Step

1. **Fine-tuning Dataset**
Domain adaptation splits released
2. **Multimodal Extension**
RF measurements & simulation plots integration
3. **Industry Collaboration**
Partnership with 6G research alliances

Future Vision

AI-Assisted Wireless Design

Enabling LLMs to become reliable partners
in next-generation wireless system
development

Community Building

Open benchmark fosters collaborative
research in AI + Wireless

Thank You!

 Questions?



<https://lixin.ai/WirelessMathBench>

Join us in advancing AI-assisted wireless system design